

Exercise 4 - Excited states and spectra

TD-DFT for the first five singlet excited states at the four optimised structures of Exercise 1, with the same functional, basis and solvent model as the optimisation: ! RKS <PBE | B3LYP> def2-TZVP def2/J TightSCF (RIJCOSX for B3LYP, CPCM(Water) for the solvated runs) and %tddft NRoots 5 TDA false end, i.e. full linear-response TD-DFT rather than the Tamm–Dancoff approximation. The ground state is closed-shell, so a restricted reference was used; it gives the same ground state as the unrestricted one of Exercise 1 but pure singlet excited states.

Excitation energies of the first five excited states

Table 1. TD-DFT/def2-TZVP singlet excitations: energy E , wavelength λ , oscillator strength f and the orbital transitions with weight ≥ 0.15 (HOMO = orbital 32, LUMO = orbital 33).

Method	State	E (eV)	λ (nm)	f	Main transitions (weight)
PBE (gas)	S ₁	4.656	266.3	0.0401	HOMO→LUMO (0.81), HOMO-1→LUMO+1 (0.18)
	S ₂	5.175	239.6	0.0001	HOMO→LUMO+2 (0.99)
	S ₃	5.478	226.3	0.0244	HOMO→LUMO+1 (0.63), HOMO-1→LUMO (0.32)
	S ₄	5.865	211.4	0.0000	HOMO-1→LUMO+2 (0.98)
	S ₅	6.129	202.3	0.2100	HOMO-1→LUMO (0.43), HOMO-1→LUMO+1 (0.25)
B3LYP (gas)	S ₁	4.985	248.7	0.0447	HOMO→LUMO (0.79), HOMO-1→LUMO+1 (0.20)
	S ₂	5.690	217.9	0.0002	HOMO→LUMO+2 (0.97)
	S ₃	5.749	215.7	0.0154	HOMO→LUMO+1 (0.60), HOMO-1→LUMO (0.38)
	S ₄	6.339	195.6	0.0000	HOMO-1→LUMO+2 (0.96)
	S ₅	6.667	186.0	0.0009	HOMO→LUMO+3 (0.98)
PBE (water)	S ₁	4.643	267.0	0.0585	HOMO→LUMO (0.83), HOMO-1→LUMO+1 (0.16)
	S ₂	5.267	235.4	0.0002	HOMO→LUMO+2 (0.99)
	S ₃	5.449	227.5	0.0377	HOMO→LUMO+1 (0.64), HOMO-1→LUMO (0.31)
	S ₄	5.936	208.9	0.0000	HOMO-1→LUMO+2 (0.98)
	S ₅	6.018	206.0	0.3226	HOMO-1→LUMO (0.54), HOMO→LUMO+1 (0.19)
B3LYP (water)	S ₁	4.974	249.3	0.0647	HOMO→LUMO (0.81), HOMO-1→LUMO+1 (0.18)
	S ₂	5.720	216.7	0.0216	HOMO→LUMO+1 (0.61), HOMO-1→LUMO (0.37)
	S ₃	5.762	215.2	0.0003	HOMO→LUMO+2 (0.97)
	S ₄	6.390	194.0	0.5322	HOMO-1→LUMO (0.58), HOMO→LUMO+1 (0.34)
	S ₅	6.395	193.9	0.0068	HOMO-1→LUMO+2 (0.95)

Influence of the solvent

Table 2. Excitation energies (eV) in the gas phase and in water, matched by their main transition (the state number is given in brackets, because the order changes), and the solvent shift $\Delta E = E(\text{water}) - E(\text{gas})$. “—”: not among the first five states.

Main transition	PBE gas	PBE water	ΔE	B3LYP gas	B3LYP water	ΔE
HOMO→LUMO	4.656 (S ₁)	4.643 (S ₁)	-0.013	4.985 (S ₁)	4.974 (S ₁)	-0.011
HOMO→LUMO+2	5.175 (S ₂)	5.267 (S ₂)	+0.092	5.690 (S ₂)	5.762 (S ₃)	+0.072
HOMO→LUMO+1	5.478 (S ₃)	5.449 (S ₃)	-0.029	5.749 (S ₃)	5.720 (S ₂)	-0.029
HOMO-1→LUMO+2	5.865 (S ₄)	5.936 (S ₄)	+0.071	6.339 (S ₄)	6.395 (S ₅)	+0.056
HOMO-1→LUMO	6.129 (S ₅)	6.018 (S ₅)	-0.111	—	6.390 (S ₄)	—
HOMO→LUMO+3	—	—	—	6.667 (S ₅)	—	—

Water changes the excitation energies only slightly: by at most 0.11 eV for the states that appear in both phases. The lowest state, S_1 , is the HOMO→LUMO excitation for all four methods and moves by only −0.013 eV with PBE and −0.011 eV with B3LYP (266.3 → 267.0 nm and 248.7 → 249.3 nm). The states that end in the HOMO→LUMO+1 and HOMO−1→LUMO configurations are lowered by water, whereas those ending in LUMO+2 are raised by 0.06–0.09 eV, so S_2 and S_3 swap order for B3LYP. The effect on the intensities is larger than on the energies: the oscillator strength of S_1 grows from 0.040 to 0.058 (PBE) and from 0.045 to 0.065 (B3LYP). The choice of functional matters much more than the solvent: B3LYP places every state 0.3–0.5 eV above PBE.

UV spectra

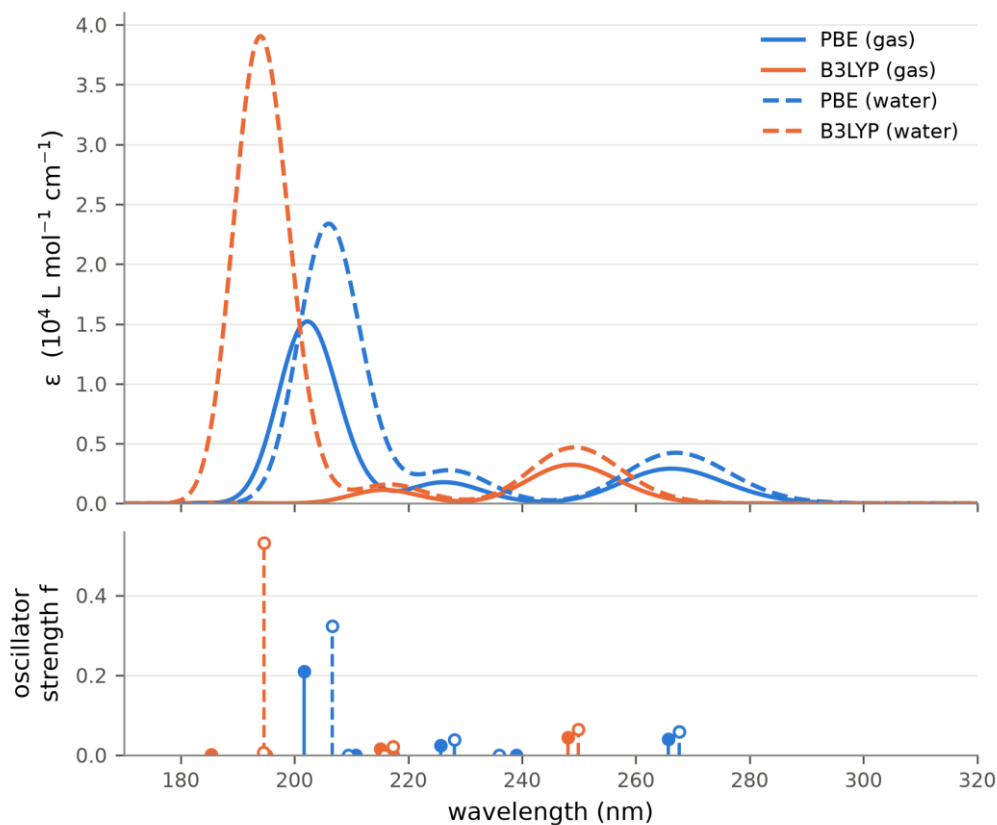


Figure 1. Simulated UV spectra of 2-chlorophenol. Top: each transition broadened by a Gaussian of FWHM 3000 cm^{-1} ($\approx 0.37 \text{ eV}$) in wavenumber, $\epsilon = \sum 2.174 \times 10^8 f_i / \text{FWHM} \cdot \exp[-2.773 (\tilde{\nu} - \tilde{\nu}_i)^2 / \text{FWHM}^2]$. Bottom: oscillator strengths (filled: gas, open: water; gas and water sticks are offset by $\pm 0.6 \text{ nm}$ for visibility). Only five states were computed, so the spectra are incomplete below about 200 nm.